

Diagrammatic equation showing the simplification of a vertex with three legs and a loop into a vertex with three legs.

Left side (Vertex with loop): A vertex (black dot) with three legs and a loop. The legs are labeled $j_1 m'_1$ (top), $j_2 m'_2$ (middle), and $j_3 m'_3$ (bottom). The loop is a vertical chain of four squares. The top leg has a '+' sign above it. The middle leg has a '+' sign above the loop. The bottom leg has a '+' sign below the loop. The vertex is preceded by a '-' sign.

Right side (Vertex without loop): A vertex (black dot) with three legs labeled $j_1 m'_1$ (top), $j_2 m'_2$ (middle), and $j_3 m'_3$ (bottom). The vertex is preceded by a '-' sign.

The equation is represented as:

$$- \text{Vertex with loop} = - \text{Vertex without loop}$$